

Relive the CERN70 Community Event



The main stage featured speeches, musical performances and iconic moments from CERN's 70-year history. (Images: Max Brice, Mike Struik and Marina Cavazza/CERN)

"It is you, and everyone who came before you, who have written CERN's history, and it is you who are continuing to shape its future," said CERN Director-General, Fabiola Gianotti, to the CERN community on Tuesday, 17 September, as we came together to celebrate the Laboratory's 70th birthday.

About 8000 people, including members of the personnel, contractors, students, visiting scientists, friends and family participated in the event on the CERN Meyrin site. The evening included music, speeches and a special birthday cake parade.

Hosts Paola Catapano and Stephan Petit welcomed the community ([video](#) 0:00:00 – 0:05:59) before being joined onstage by special guests ([video](#) 0:05:59 – 0:42:01), including children from CERN's Jardin des Particules and crèche alumni; capacitive touchscreen pioneer, Bent Stumpe; Director for Research and Computing and former PhD student, Joachim Mnich; CERN alumni; Beamline for Schools students; a trailer by filmmaker Cedric Klapisch and more.

A Word from Mar Capeans

Latest improvements to life at CERN

From the new CERN Community Support Centre to the CERN Campus App, find out how the renovations and innovations of the Site and Civil Engineering department are helping to improve life at CERN

Contents

News.....p.1-16

Relive the CERN70 Community Event

CERN70 official event on 1 October: your questions answered

Accelerator Report: Preparing the LHC and its injector complex for the 2024 lead-ion run

CERN70: Superconductors accelerate progress

LHC experiments at CERN observe quantum entanglement at the highest energy yet
CMS experiment at CERN weighs in on the W boson mass

ATLAS probes Higgs interaction with the heaviest quarks

Take a virtual walk on the wild side

How can physicists make particle accelerators more efficient?

A Nobel ninetieth

Detector now collecting data in Japan after upgrade at CERN

Computer Security: AI attacks

Official News.....p.16-18

Authorisation to operate CERN lifting equipment

Voluntary insurance available for staff members, fellows and graduates

Announcements.....p.18-19

Upcoming events

Sign your child up for the *Futur en tous genres* programme and share your passion for your work!

Closure of the tunnel linking the CERN sites on 23 and 24 September

The smart(phone) way to celebrate CERN70



A musical journey through CERN's history began with concerts by members of the CERN Music club ([video](#) 00:42:05 – 02:00:04) including the Oliver Street Band, the Canettes Blues Band – who made a memorable entrance by car, Miss Proper and The Moving Targets, and Diracula. Following them was a concert by Les Horribles Cernettes (pictured here) ([video](#) 02:00:04 – 02:51:00).

The speech by the CERN Director-General to the community ([video](#) 02:51:00 – 03:01:20) included these words:

“The history of CERN is adorned with great discoveries, observations and measurements, made in a succession of extraordinary machines and countless technological breakthroughs. These successes are those of a community, the CERN community, which includes staff, users, fellows, students, graduates and everyone who works here in one way or another. This community, with its competence, expertise, creativity, passion and determination, is CERN's strongest asset.”

Fabiola Gianotti, CERN Director-General



CERN Director-General, Fabiola Gianotti joined the hosts, offering cake to attendees as CERN's

gigantic 70th birthday cake was paraded through the crowd ([video](#) 03:01:20 – 03:11:04), accompanied by a rousing chorus of “Happy Birthday”. The cake's composition was [inspired by the Universe](#).



Exclusive musical performances included the *Orchestre des Nations* ([video](#) 03:11:04 – 03:30:47), who were joined by John Helliwell and Carl Verheyen ([video](#) 03:30:47 – 03:43:00). John Helliwell and the Carl Verheyen Band then performed together ([video](#) 03:43:00 – 04:58:51) until the *Orchestre des Nations* rejoined them for songs by Supertramp including “The Logical Song” and “Give a Little Bit” ([video](#) 04:58:51 – 05:16:26).



(Image: Prakhar Katyayan/CERN)

Beyond the main stage, activities included magic shows, dancing, games and CERN-themed temporary tattoos.



A dedicated exhibition of [vintage photos](#) was prepared for the event. Some of these images are also featured in the latest [CERN Courier](#).



Almost 100 volunteers helped the event to run as smoothly as possible. Many thanks to them and to all the teams at CERN who were involved in making the event a success.

It was certainly a night to remember.

See more of the [photo](#) and [video](#) highlights (CERN login required).

Internal Communication

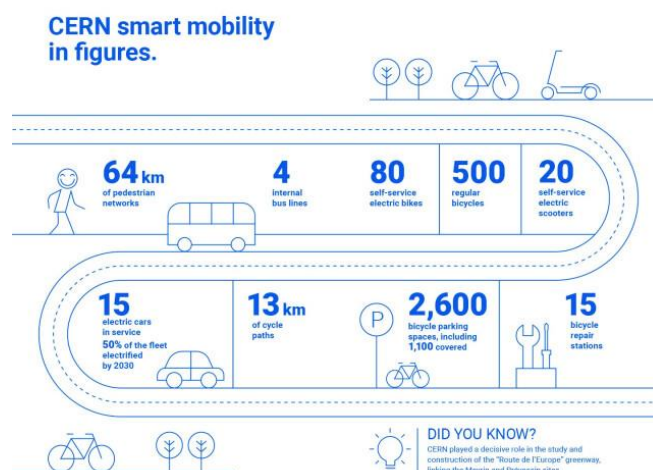
A Word from Mar Capeans

Latest improvements to life at CERN

From the new CERN Community Support Centre to the CERN Campus App, find out how the renovations and innovations of the Site and Civil Engineering department are helping to improve life at CERN

As we come together to celebrate 70 years of CERN, we also acknowledge that parts of the CERN sites are really showing their age. As Head of the Site and Civil Engineering (SCE) department, this means overseeing a department that is constantly striking a balance between renovation and innovation.

At the SCE plenary meeting on 3 September, we took stock of the work done so far in 2024. Quite frankly, the results are fantastic. The video below shows the new buildings and renovations on the CERN sites this year alone, from the [Prévessin Data Centre](#) to the [Main Building](#). It also highlights the new [CERN Community Support Centre](#) in Building 33, which is now the first line of support for all the CERN services and activities that were previously housed in Building 55, from registrations and access cards to biometry registration, dosimeter distribution and [more](#).



Graphic from the new [smart mobility brochure](#) with information about bicycles, shuttles, car sharing and more. (Image: CERN)

In addition, the [CERN Campus App](#) goes from strength to strength. [Launched](#) last year by the SCE and Finance and Administrative Processes (FAP) departments, it provides the community with maps, the phonebook, news, restaurant information and much more. It was also used during this week's CERN70 Community Event by more than 1500 users to receive site information in real time. Its emergency push notifications are tested on the first Wednesday of each month, and to date about 10 000 phones have received these messages. If you haven't downloaded the app yet, do so [here](#) and make sure that push notifications are enabled.

Collaboration with CERN's Occupational Health & Safety and Environmental Protection (HSE) unit has also led to increased awareness of our [recuperation schemes](#) and the new [biodiversity](#)

[virtual walk](#). Earlier this year, CERN's [Mobility Plan](#) was approved and is now helping to promote sustainable and [smart](#) transport methods. While I stood at this week's CERN70 Community celebration, surrounded by representatives of more than 100 countries, I couldn't help but think what a remarkable place we work in. I'm proud to work at CERN and proud of my colleagues who strive to constantly improve life on the CERN campus. Thank you all.

Watch the video overview of the new buildings and renovations on the CERN sites in 2024: <https://videos.cern.ch/record/2300589>.

*Mar Capeans
Head of the Site and Civil Engineering department*

CERN70 official event on 1 October: your questions answered

What is happening? Should I avoid the Meyrin site? We answer these questions and more about the CERN70 official event

CERN celebrates its 70th anniversary on 29 September 2024 and there have been events and activities all year, building up to the official CERN70 event on 1 October.

1. **EVENT: What is happening on 1 October at CERN?**

On Tuesday, 1 October, an official high-level ceremony will welcome political leaders from Member and Associate Member States, as well as from Observers. With speeches from the Director-General and the President of Council, as well as from Heads of State or Government, the event will showcase the Laboratory's rich past and bright future. A series of short films will highlight significant milestones in CERN's history, and the event will close with a performance from the Orchestre de la Suisse Romande (OSR).

2. **WEBCAST AND SOCIAL MEDIA: How can I be part of this event?**

Due to capacity constraints, the auditorium cannot accommodate the CERN community. There is only enough space for the officially invited delegations and external guests.

The event will nonetheless be broadcast live via [CERN webcast](#), starting at 3 p.m. and scheduled to finish at 4.30 p.m.

The webcast will also be retransmitted to the **Main Auditorium (Building 500)** on the Meyrin site and the **BE auditorium (774-R-013)** on the Prévessin site, should you wish to watch the event alongside your colleagues.

You can also join the conversation on social media, tagging **@CERN** and using the hashtag **#CERN70**.

3. **SITE INFORMATION: Which services and car parks will be unavailable?**

The **CERN Community Support Centre (CCSC) in Building 33** will be **unavailable** to the CERN community **from 11 a.m. to 3.30 p.m.** as it will be exclusively registering press and invited CERN70 guests. However, the **Service Desk** will continue to provide support:

- by email service.desk@cern.ch, access.registration@cern.ch
- by phone 77777 (+41 22 76 77777)
- or through the [service portal](#)

CERN Science Gateway, the Globe of Science and Innovation and IdeaSquare will be closed. Access

to this area will be **strictly controlled** by police and security services.

Access to the **ATLAS** site (LHC Point 1) will be **severely disrupted** due to the convoys of delegations, police and security services.

Access to the **Complexe Sportif de Maisonnex** may also be disrupted.

The **visitor car park** next to CERN Science Gateway will be **closed** on 30 September and 1 October. The **Les Cerisiers car park** (next to Buildings 4 and 5) will be **closed**, as will **part of the Greinacher car park** (under the power lines). These car parks will be used by the invited delegations.

4. NEWCOMERS: What about the newcomers arriving at CERN on 1 October?

More than **150 newcomers** will start their CERN contracts on this unique day. The CERN Community Support Centre at Building 33 will be open from 7 to 11 a.m. and from 3.30 to 6.30 p.m. to finalise the registration formalities that will have already begun online. Newcomers are **recommended to arrive early**. Their **departmental administrative officers** and **supervisors** will then welcome them into the team. Because the Globe will be unavailable, the Welcome Session will exceptionally take place on 2 October in the Council Chamber.

5. PLANNING: Should I avoid the CERN Meyrin site that day?

If you are expecting a newcomer, please make the necessary arrangements to welcome them.

If **teleworking** is compatible with your functions, you may prefer to work from home to avoid traffic disruptions. In that case, please submit an EDH absence request, which will need to be signed by your supervisor.

If being on site is essential, please **anticipate traffic disruptions** and **plan your journey** accordingly.

On **Tuesday, 1 October**, major traffic disruptions are expected on the **roads around the Meyrin site** from **10.30 a.m. to 12 noon** and from **4.30 to 6 p.m.** including the Route de Meyrin and roads between the CERN Meyrin site and Geneva airport.

These disruptions may also affect **bus 68** and **tram 18** to and from CERN.

This official event is set to be a moment to remember, as invited representatives from across CERN's Member States, Associate Member States and Observers come together to celebrate this historic milestone.

Internal Communication

Accelerator Report: Preparing the LHC and its injector complex for the 2024 lead-ion run

The physics run for the SPS North Area will start on 4 November, followed by the LHC lead-ion physics run on 6 November, and the PS East Area on 11 December

In the early morning of 13 September, the LHC reached a significant milestone, surpassing 100 fb⁻¹ of integrated luminosity delivered to ATLAS and CMS in a single year – a record – and the figure is still rising. Moreover, LHCb's integrated luminosity target of 8.5 fb⁻¹ was reached one day later. Of course, this does not mean that we will cease delivering collisions to the LHCb experiment – the LHC team will do its utmost to further increase the size of LHCb's dataset.

While the LHC continues to provide proton-proton collisions to the experiments, the injector chain is,

simultaneously, providing protons to the fixed target experiments. It is also preparing for the lead-ion physics run, not only in the LHC but also in the SPS North Area and the PS East Area.

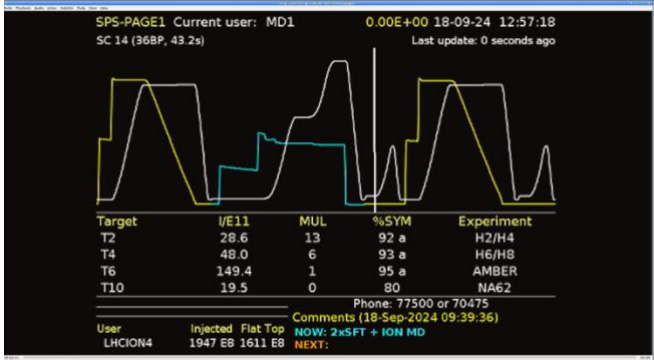
After a brief test with magnesium ions from April to May this year, Linac3 was converted back to produce lead ions. The original plan was to inject lead ions into the LEIR machine on 5 August. However, due to a problem with the high-voltage power supply in a 14.5 GHz generator that feeds the Linac3 source, this was delayed. Following repairs to the generator, Linac3 restarted on 19

August and, that same day, the first lead ions were available for LEIR.

Because of this two-week delay, the LEIR beam commissioning schedule had to be adjusted. Studies to improve LEIR's performance were cancelled and a fast-tracked plan was introduced to ensure that lead-ion beams were ready in time to be delivered to the SPS.

On 17 September, as planned, the first lead ions were successfully injected and circulating in the SPS. Since the SPS usually operates with a high beam duty cycle and has little flexibility for unscheduled additions, special commissioning blocks for the lead ions were included in the injector schedule for the start of the year.

On 18 and 19 September, from 8 a.m. to 6 p.m., a short ion cycle (14.4 seconds) was added to the SPS supercycle (see image), allowing the SPS team to perform the first part of the lead-ion beam commissioning in the accelerator. For the next steps, a longer ion cycle (57.6 seconds) is required. However, this long cycle would significantly extend the SPS supercycle, reducing the efficiency for both the protons and the ion beam users. To manage this efficiently, six dedicated 10-hour commissioning blocks have been scheduled between 1 and 30 November, during which the long ion cycle will be the sole focus of the SPS supercycle. This will allow the team to set up the final lead-ion beam for the LHC, including [the slip-stacking process](#).



The current SPS supercycle, which repeats every 43.2 seconds, starts with a proton cycle for the SPS North Area, followed by short lead-ion cycles. Another proton cycle then ends the supercycle. The white lines represent the energy of the beam, the yellow line the number of protons and the blue line the number of lead ions. (Image: CERN)

Finally, at least two versions of the lead-ion beams will be prepared in the SPS: one of 57.6 seconds for the LHC, incorporating the slip-stacking process; and another for the users in the SPS North Area, which will involve slow extraction but no slip-stacking and will be 25.2 seconds long. The four-week physics run for the SPS North Area users is scheduled to start on 4 November, followed by the LHC lead-ion physics run on 6 November, which will last 2.5 weeks. The lead-ion physics programme in the PS East Area is set to start on 11 December.

Due to the fast-tracked commissioning process in LEIR, valuable time was lost for studying and improving machine performance, as well as for addressing issues encountered during the 2023 lead-ion run. However, the LEIR team will use all available time to recover as much as possible of the lost study time between the six dedicated commissioning blocks.

Rende Steerenberg

CERN70: Superconductors accelerate progress

Part 17 of the [CERN70](#) feature series. Lucio Rossi led the group responsible for superconducting magnets for the LHC

The [LHC](#), the largest superconducting machine in the world, demonstrates how particle physics and CERN have been a driving force in the development of superconductors.

Superconductivity quickly emerged as a very useful property for high-energy physics. Since superconductors lose all electrical resistance below a certain, very low temperature, they are able to transport very high

electrical currents, which makes it possible to produce magnets that generate strong magnetic fields, as well as powerful accelerating cavities.

Detectors were equipped with the first superconducting magnets from the 1960s onwards. In 1965, an experiment at CERN's [Synchrocyclotron](#) began testing a small superconducting coil that generated a magnetic field of 4.2 teslas. A few years later, in 1973, the Big European Bubble Chamber (BEBC) was fitted with a huge superconducting magnet of over three metres in diameter, a record at the time.



The superconducting coil of the large European BEBC bubble chamber in 1974, already the largest superconducting magnet at the time. (Image: [CERN](#))

Detectors have progressively been equipped with increasingly powerful magnets, up to the scale of the giant ones in use at the LHC. [CMS](#) boasts the largest superconducting solenoid in the world and [ATLAS](#) the largest toroidal magnet, made up of 25-metre-long coils.

The magnets that are destined for use in accelerators must be produced in their dozens, their hundreds or even their thousands, which explains why it took until the 1980s for the first superconducting accelerators to come into service. The Tevatron collider in the USA, which came online in 1983, was the first large-scale application of these magnets. In 1989, the [Large Electron-Positron Collider \(LEP\)](#) at CERN was equipped with a few superconducting magnets designed to focus the bunches of particles before they collided. But, above all, superconducting accelerating cavities were installed in the LEP from 1996 onwards: more than 280 in number, they were capable of generating more intense accelerating fields to increase the energy of the collider.

Even before the LEP came into service, the teams at CERN were already working on its successor, a proton collider to be installed in the same tunnel, based on superconducting magnets generating fields of more than 8 teslas. The epic technological and industrial task of creating the largest superconducting facility in the world would take 25 years to complete. This collider, the LHC, contains more than 9000 niobium-titanium superconducting magnets, including 1232 15-metre-long dipole magnets that each weigh 35 tonnes. Of the

accelerator's 27 km circumference, 23 km are cooled with superfluid helium to 1.9 degrees above absolute zero (-271 °C).

The LHC has also prompted work on high-temperature superconductors, which do not have to be cooled to such glacial temperatures. The LHC current leads – the connections in which the current passes from the ambient temperature to the temperature of the liquid helium in the magnets – are made of these superconductors, which are ceramic and very complicated to deploy.

The [High-Luminosity LHC](#), the major upgrade of the LHC that is currently being built, also brings its own share of innovations in the field of superconductors. It will be equipped with magnets made from a superconducting niobium-tin compound, which will produce more intense magnetic fields of 12 teslas. It will also use electrical connections made from intermediate-temperature superconductors, which can operate at 25 kelvins (-248°C). The CERN teams are continuing their work on high-temperature superconductors, the Holy Grail for experts in this field, which, in the longer term, could open up unprecedented possibilities for fundamental physics and have numerous applications for the benefit of society in general.

Recollections

Lucio Rossi began his career in plasma physics and moved into the field of applied superconductivity for accelerators in the early 1990s, heading the INFN-LASA team that had just begun to work on the first prototype superconducting dipole magnet for the LHC. In 2001, he became the leader of CERN's Superconducting Magnets and Cryostats group, which was responsible for developing and launching the industrial production of the superconducting magnets needed for the LHC. From 2010 to 2020, he was the leader of the High-Luminosity LHC project.



To motivate his team, Lucio Rossi (sitting third from left) installed an electronic display panel displaying the number of superconducting dipole magnets still to be manufactured out of the total of 1232. Every 154 magnets (every octant of the LHC ring) a small celebration was organised. The display showed zero in November 2006. (Image: CERN)

magnet has been very successful and that first dipole was used until recently by the CAST experiment – you could still read the words “LHC DIPOLE PROTOTYPE CERN-INFN 1” on it.

A few years later, in 2000, I received a phone call from Lyn Evans, then leader of the LHC project, who asked me if I would consider leading CERN's team in charge of superconducting magnets. It took me approximately one microsecond to say yes! Then started this incredible adventure of building the most complex accelerator in the world, using the most complex technology. We had to produce 300 000 km of superconducting wire for the LHC superconducting dipole and quadrupole magnets and for the several thousand corrector magnets, and to do so we had to collaborate with industry. That was not easy. Moving from our complex superconducting dipole magnet prototypes to industrialisation was a challenge, in particular when it came to getting the level of quality that we wanted from the companies. We had to make compromises, because sometimes best is the enemy of good. It was difficult, but we did it.

“The INFN-LASA team delivered the first LHC prototype magnet to CERN in 1994. This type of



Some of the 1232 dipole magnets for the LHC stored in 2003 in the car park of the magnet test hall. The large-scale manufacture of superconducting magnets for the LHC has enabled CERN and industry to develop unique expertise in the field of superconductivity. (Image: Peter Ginter/[CERN](#))

Producing the magnets was a real challenge; assembling them and making them work together was yet another one. The difficulty with the LHC is that everything is in series, so if you have a single weak link, the chain breaks. And that's [what happened on 19 September 2008](#). One single electrical connection between two adjacent magnets – out of 10 000 such connections – failed, leading to extensive collateral damage and a one-year delay. I have to assume that the mistake was made by us, the engineers, the designers, because we had thought of a system that was not robust against faulty workmanship.

We learnt from this incident collectively. I think this is the strength of CERN: at that moment, I could feel that CERN was united. Afterwards, the LHC was resurrected stronger than before, and we got the Higgs boson.

When the LHC restarted, in November 2009, it was really a triumphal march. I was very moved to see

the LHC restarting, to see the machine working well. We had done a good job. That was a glorious day.

In 2010, we then started to think about the future of CERN, and I proposed the High-Luminosity LHC project. This was an important project for me, as the HL-LHC sort of closed the loop for me personally, so I took on the role of HL-LHC Project Coordinator in 2010. From then on, we flew... but not without some difficulties, as one can expect in such a project, the biggest one probably being COVID-19. We also had some difficulties with the magnets: for the HL-LHC, we decided to build new types of magnets, based on the new niobium-tin superconducting technology, while the LHC's magnets are based on the niobium-titanium technology.

We thought that we would set up the technology on a short magnet, a 1.5-metre-long magnet, and then just extend it to a 7-metre-long magnet. It was like that for the LHC. But we learnt the hard way that it was not so easy with the niobium-tin technology, as the coils are much more fragile; that's why we faced [these challenges, in particular with the HL-LHC 11 T magnets](#). We also learnt from failures, and we now have successful long magnets for the inner triplet quadrupoles, a key element of the HL-LHC, but it took a lot of time. That was a bumpy ride but it was necessary. The HL-LHC will be a real test of the niobium-tin technology, to help make the jump to future colliders.

CERN is here to make things that are truly difficult, to dream what is not yet possible. I'm convinced of this, otherwise it wouldn't be CERN."

LHC experiments at CERN observe quantum entanglement at the highest energy yet

The results open up a new perspective on the complex world of quantum physics

Quantum entanglement is a fascinating feature of quantum physics – the theory of the very small. If two particles are quantum-entangled, the state of one particle is tied to that of the other, no matter how far apart the particles are. This mind-bending phenomenon, which has no analogue in classical physics, has been observed in a wide variety of

systems and has found several important applications, such as quantum cryptography and quantum computing. In 2022, the [Nobel Prize in Physics](#) was awarded to Alain Aspect, John F. Clauser and Anton Zeilinger for groundbreaking experiments with entangled photons. These experiments confirmed the predictions for the

manifestation of entanglement [made by the late CERN theorist John Bell](#) and pioneered quantum information science.



Artist's impression of a quantum-entangled pair of top quarks. (Image: CERN)

Entanglement has remained largely unexplored at the high energies accessible at particle colliders such as the [Large Hadron Collider](#) (LHC). In an [article](#) published today in *Nature*, the [ATLAS](#) collaboration reports how it succeeded in observing quantum entanglement at the LHC for the first time, between fundamental particles called top quarks and at the highest energies yet. First reported by ATLAS in [September 2023](#) and since confirmed by [two observations](#) made by the [CMS](#) collaboration, this result has opened up a new perspective on the complex world of quantum physics.

"While particle physics is deeply rooted in quantum mechanics, the observation of quantum entanglement in a new particle system and at much higher energy than previously possible is remarkable," says ATLAS spokesperson Andreas Hoecker. "It paves the way for new investigations into this fascinating phenomenon, opening up a rich menu of exploration as our data samples continue to grow."

The ATLAS and CMS teams observed quantum entanglement between a top quark and its [antimatter](#) counterpart. The observations are based on a [recently proposed method](#) to use pairs of top quarks produced at the LHC as a new system to study entanglement.

The top quark is the heaviest known fundamental particle. It normally decays into other particles before it has time to combine with other quarks, transferring its spin and other quantum traits to its decay particles. Physicists observe and use these

decay products to infer the top quark's spin orientation.

To observe entanglement between top quarks, the ATLAS and CMS collaborations selected pairs of top quarks from data from proton–proton collisions that took place at an energy of 13 teraelectronvolts during the second run of the LHC, between 2015 and 2018. In particular, they looked for pairs in which the two quarks are simultaneously produced with low particle momentum relative to each other. This is where the spins of the two quarks are expected to be strongly entangled.

The existence and degree of spin entanglement can be inferred from the angle between the directions in which the electrically charged decay products of the two quarks are emitted. By measuring these angular separations and correcting for experimental effects that could alter the measured values, the ATLAS and CMS teams each observed spin entanglement between top quarks with a statistical significance larger than [five standard deviations](#).

In its [second study](#), the CMS collaboration also looked for pairs of top quarks in which the two quarks are simultaneously produced with high momentum relative to each other. In this domain, for a large fraction of top quark pairs, the relative positions and times of the two top quark decays are predicted to be such that classical exchange of information by particles traveling at no more than the speed of light is excluded, and CMS observed spin entanglement between top quarks also in this case.

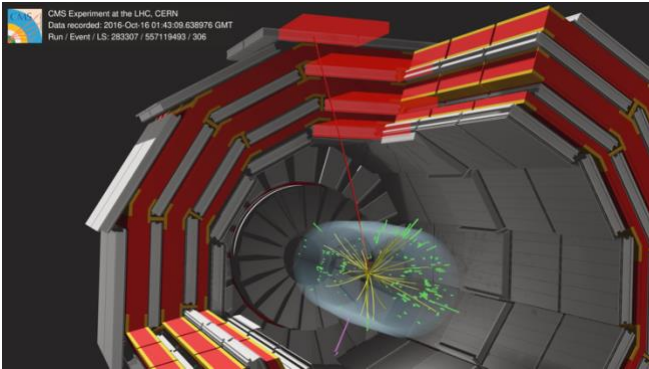
"With measurements of entanglement and other quantum concepts in a new particle system and at an energy range beyond what was previously accessible, we can test the [Standard Model](#) of particle physics in new ways and look for signs of new physics that may lie beyond it," says CMS spokesperson Patricia McBride.

Read more:

- [ATLAS Nature paper](#)
- [CMS first study](#)
- [CMS second study](#)

CMS experiment at CERN weighs in on the W boson mass

The eagerly awaited result is the most precise measurement of the W mass made at the LHC so far, and is in line with the prediction from the Standard Model of particle physics



CMS candidate collision event for a W boson decaying into a muon (red line) and a neutrino that escapes detection (pink arrow). (Image: CMS/CERN)

The [CMS](#) experiment at CERN is the latest to weigh in on the mass of the [W boson](#) – an elementary particle that, along with the [Z boson](#), mediates the weak force, which is responsible for a form of radioactivity and initiates the nuclear fusion reaction that powers the Sun.

At a [seminar](#) held at CERN today, the CMS collaboration reported how it has analysed proton–proton collision data from the second run of the [Large Hadron Collider](#), the Laboratory’s flagship particle accelerator, to make its first mass measurement of this fundamental particle.

The [result](#) is the most precise measurement of the W mass made so far at the LHC, and is in line with the prediction from the [Standard Model](#) of particle physics and with all previous measurements, except the measurement from the CDF experiment at the former proton–antiproton [Tevatron](#) collider at Fermilab.

In the Standard Model, the W mass is related closely to the strength of the interaction unifying the electromagnetic and weak forces and to the masses of the [Higgs boson](#) and the [top quark](#), which constrain its value to 80353 million electronvolts (MeV) within an uncertainty of 6 MeV.

Measuring the W boson mass with high precision therefore makes it possible to test whether or not these properties all align in a way that is consistent with the Standard Model. If they don’t, the cause could be new physics phenomena such as new particles or interactions.

Since its discovery at CERN about [40 years ago](#), the W boson has had its mass measured ever more precisely by several collider experiments, including [ATLAS](#) and [LHCb](#) at the LHC. In 2022, a surprisingly high value of its mass measured by the CDF experiment plunged the particle into a “[midlife crisis](#)”. The [CDF W boson mass](#), 80433.5 MeV with an uncertainty of 9.4 MeV, differed significantly from the Standard Model prediction and from the other experimental results, calling for more studies.

In 2023, the [ATLAS](#) collaboration, which provided its first W boson mass measurement in 2017, [released](#) an improved measurement based on a reanalysis of proton–proton collision data from the first run of the LHC. This improved result, [80366.5 MeV with an uncertainty of 15.9 MeV](#), lined up with all previous measurements except the CDF measurement, which remains the most precise to date, with a precision of 0.01%.

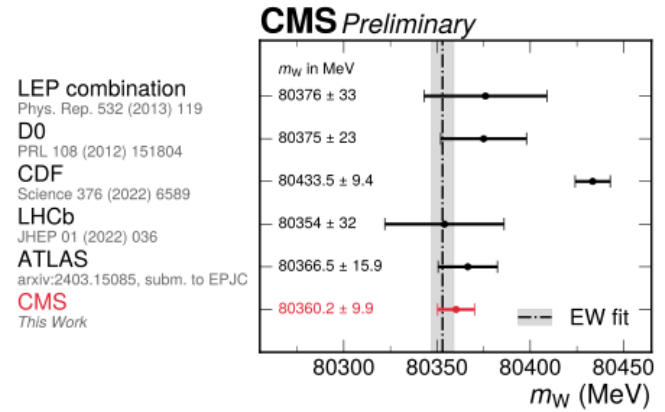
The CMS experiment has now contributed to this global endeavour with its first W boson mass measurement. The keenly anticipated result, 80360.2 with an uncertainty of 9.9 MeV, has a precision comparable to that of the CDF measurement and is in line with all previous measurements except the CDF result.

“The wait for the CMS result is now over. After carefully analysing data collected in 2016 and going through all the cross checks, the CMS W mass result is ready,” says outgoing CMS spokesperson Patricia McBride. “This analysis is the first attempt to measure the W mass in the harsh collision environment of the second running period of the LHC. And all the hard work from the team has resulted in an extremely precise W mass measurement and the most precise measurement at the LHC.”

“W mass measurements are very challenging, involving delicate measurements and theoretical modelling of the production of the W boson and its decay into a lepton (here, a muon) and a neutrino that escapes detection,” explains incoming CMS spokesperson Gautier Hamel de Monchenault. “By exploiting the ability of the CMS

detector to measure muons with high precision and using the latest and most advanced theoretical ingredients, some of which were tested by a cross-checking analysis, we attained this record level of precision.”

The result showcases once again the exceptional performances of the LHC and its detectors, which continue to push the precision frontier and put the Standard Model and its extensions to ever more stringent tests. Further data from the ongoing third run of the collider and from its upgrade, the [High-Luminosity LHC](#), is poised to push this frontier further.



Measurements of the W boson mass reported over the years by experiments worldwide. The CMS measurement is the most precise one made at the LHC so far, and has a precision comparable to that of the CDF measurement. (Image: CMS/CERN)

ATLAS probes Higgs interaction with the heaviest quarks

With its updated analysis methods, the ATLAS collaboration has significantly improved the precision of its measurements of Higgs boson interactions

A central aim of the ATLAS Higgs physics programme is to measure, with increasing precision, the strength of interactions of the Higgs boson with elementary fermions and bosons. According to the theory of electroweak symmetry breaking, these interactions are responsible for generating the masses of the particles. The interaction strengths can be determined by precisely measuring the Higgs boson’s production via and decay into the relevant particles.

At the recent International Conference on High-Energy Physics (ICHEP) 2024, the ATLAS collaboration presented improved measurements of the strength of Higgs boson interactions with the three heaviest quarks: top, bottom and charm. The new results are based on a reanalysis of LHC Run 2 data taken in the years 2015–2018 with significantly enhanced analysis methods, including improved jet tagging.

But what are jets and why do they need to be tagged? When the Higgs boson decays into a pair of quarks, each quark fragments, creating a collimated spray of particles (mostly hadrons) that can be observed in the detector. The aim of jet tagging is to determine which type (or “flavour”) of quark produced a given jet through detailed analysis of the jet’s properties. With new bespoke jet (or “flavour”) tagging techniques for charm and

bottom quarks, ATLAS researchers managed to significantly increase the sensitivity of their analyses. Together with other analysis improvements, they increased sensitivity to $H \rightarrow b\bar{b}$ and $H \rightarrow c\bar{c}$ decays by 15% and a factor of three, respectively.

Updated measurements of Higgs boson production in association with a W or Z boson and decays into a pair of bottom or charm quarks yielded the first observation of the WH , $H \rightarrow b\bar{b}$ process with 5.3σ significance and a measurement of ZH , $H \rightarrow b\bar{b}$ with 4.9σ significance. The Higgs boson decay into c quarks is suppressed by a mass factor of 20 relative to the decay into b quarks and thus is still too rare to be observed. ATLAS sets an upper limit on the rate of the VH , $H \rightarrow c\bar{c}$ process of 11.3 times the Standard Model prediction. These results are the most precise probes of these processes to date, and they are compatible with the Standard Model.

A new measurement of Higgs boson interaction with the top quark focused on Higgs production in association with two top quarks and its subsequent decay into a pair of bottom quarks. This challenging process features a very complex final state and suffers from large backgrounds. The new analysis, which benefits from a refined understanding of the dominant background

processes involving top quarks, improved the sensitivity by a factor of two and measured a signal strength for $t\bar{t}H$, $H \rightarrow b\bar{b}$ production of 0.81 ± 0.21 , relative to the Standard Model prediction.

Further improved analysis techniques and new data from the ongoing Run 3 hold the promise of measuring these interactions with even greater

precision. These advancements in the search for $H \rightarrow c\bar{c}$ heighten anticipation for the High-Luminosity LHC (HL-LHC), where detecting this process enters the realm of feasibility.

Read more in the ATLAS briefings [here](#) and [here](#).

Piotr Traczyk

Take a virtual walk on the wild side

A new online application takes you on a journey to discover CERN's rich biodiversity

Did you know that CERN is home to a rich diversity of flora and fauna? You can now embark on a captivating virtual tour of the CERN sites to discover the many species identified in a dedicated field survey carried out in 2022. Among them you'll encounter several threatened species, providing a unique glimpse into the sites' ecological wealth.

Conducting regular surveys is crucial for maintaining CERN's biodiversity. They allow populations of flora and fauna to be monitored and help identify zones of biological interest, where concrete protection measures can be taken.

The 2022 surveys concentrated on documenting flora, amphibians, insects and birds. The inventory of amphibians revealed that two protected frog species and two protected species of the newt family live on the CERN sites. The insect-based surveys focused on Lepidoptera (butterflies) and Orthoptera (locusts, grasshoppers and crickets), which serve as indicators of environmental quality. Remarkably, 62 species of Lepidoptera and 32 species of Orthoptera were identified across the sites. The flora inventory led to the identification of 13 threatened species, as well as of two new orchid species, bringing the total number of orchid species identified on the CERN sites to 18.

This virtual walk was developed as part of CERN's comprehensive programme to prioritise and understand the environment, led by SCE in collaboration with HSE. It is powered by SCE's [Geographic Information System \(GIS\)](#), where the survey data has been stored and which allows spatial analysis and meaningful visualisations, incorporating 3D scenes from drone images and videos to provide a fully immersive user experience. Future planned developments include an online platform where everyone can contribute their own observations, enriching our collective knowledge of the biodiversity of the CERN sites. This platform will also use existing sensors and surveillance cameras to augment the inventory, leveraging AI for automatic classification. Ultimately, the resulting database will facilitate the measurement of a range of factors that impact CERN's biodiversity, including site evolution, works, seasons, weather patterns and more. This will enable us to assess the effectiveness of our actions to protect biodiversity and will deepen our understanding of the environment, helping us to better protect it.

Interested in finding out more? Click on this [link](#) and enjoy the journey!

SCE department & HSE unit

How can physicists make particle accelerators more efficient?

Using AI, machine learning and automation, the Efficient Particle Accelerators (EPA) project aims to boost the efficiency of CERN's accelerators for the high luminosity era and beyond

As particle accelerator technology moves into the high-luminosity era, the need for extreme

precision and unprecedented collision energy keeps growing. Given also the Laboratory's desire

to reduce energy consumption and costs, the design and operation of CERN's accelerators must constantly be refined in order to be as efficient as possible. To address this, the Efficient Particle Accelerators project (EPA) has been established – a team of people from different accelerator, equipment and control groups across CERN who are working together to improve accelerator efficiency.

A think-tank was set up following a 2022 workshop to plan upgrades for the [High Luminosity LHC](#) (HL-LHC), and it came up with seven recommendations on efficiency for the EPA to work on. "The idea was to look at efficiency in the broadest terms," says Alex Huschauer, engineer-in-charge of the CERN [PS](#) and member of the EPA. "We wanted a framework that could be applied to each machine in the accelerator complex." To do this, the team created nine work packages on efficiency to be deployed over the years leading up to the beginning of the HL-LHC run. "It emerged from our discussions in the efficiency think-tank that automation is the way forward," says the EPA project leader, Verena Kain. "This means using automation both in the conventional way and using AI and machine learning."

For example, AI can help physicists combat accelerator magnet hysteresis. This happens when the field of the iron-dominated accelerator magnets cannot be described by a simple mapping of current in the electromagnet to the field. If this is not taken into account, it can lead to inconsistent programmed fields and detrimental effects on beam quality, such as reducing the stability and precision of the beam's trajectory. Today, these field errors are manually tuned to correct the field, a process that takes both time and energy. "Hysteresis happens because the actual magnetic field is not defined just by the current in the power supply, but also by the magnet's history," says Kain.* "What's difficult is that we can't model it analytically – we can't work out exactly what current is needed to create the correct field for the beam in the accelerator magnet – at least not with the precision required. But AI can learn from the magnet's historical data and elaborate a precise model." The team have

done initial tests using magnets in the SPS and hope to train the AI on all CERN's accelerating magnets over the coming years.

While the experiments across the CERN accelerator complex already use automation, AI and machine learning to assist with data-taking, up until now, much of the beam and accelerator control has been done manually. "Most of the lower energy machines, like the PS, were built in an era when automation as we know it today was simply not possible." Kain continues. Another area where automation can revolutionise efficiency is in scheduling. "The different beams in the accelerator complex are produced one after the other and this has to be orchestrated so that the beam can be extracted from one machine and injected into the next at the right moment," she says. "Sometimes we have to change the schedule between 20 to 40 times a day, and it can take around 5 minutes each time. That task, currently done manually, accounts for much of the work of people in the control centre." By automating this process, control centre operators will be able to spend more time working on the beams than on scheduling.

Other areas of focus for the EPA are automated LHC filling, autopilots, automatic fault recovery and prevention, automatic testing and sequencing, automatic parameter control and optimisation. The team hopes to continue their research over the next five years, using LHC Run 3 and Long Shutdown 3 to conduct tests. "Thanks to the EPA project, for the first time we will be using AI and automation for the accelerators on a large scale," continues Huschauer. "If we can produce beams with better quality, we will be able to run the complex for less time, creating better physics data and reducing overall energy consumption."

** While this may pose a challenge for tuning the magnets, engineers at the CERN accelerators capitalise on magnet memory to train them to operate at high currents before each run. Read more about this [here](#).*

Naomi Dinmore

A Nobel ninetieth

Find out how CERN is celebrating the 90th birthday of former Director-General and Nobel Prize laureate Carlo Rubbia, and how you can be involved



A symposium to celebrate Carlo Rubbia's 90th birthday

Friday 18 October 2024

14:00 – 19:00
MAIN AUDITORIUM

Details and registration at:
<https://indico.cern.ch/event/1450340>

Speakers

Hamid Ait Abderrahim (SCK,CEN)
Amalia Ballarino (CERN)
Hesheng Chen (Chinese Academy of Sciences)
Lyn Evans (Imperial College)
Fabiola Gianotti (CERN)
Chris Llewellyn Smith (Oxford University)
Luciano Maiani (Roma University La Sapienza and INFN)
James W. Rohlf (Boston University)
Gerardus 't Hooft (Utrecht University)
Samuel Ting (MIT)
Antonio Zoccoli (Bologna University and INFN)

Award of the INFN medal,
for the 40th anniversary of the
discovery of the W and Z bosons



More information



Join us on Friday 18 October, as CERN celebrates the 90th birthday of Carlo Rubbia, CERN [Director-General](#) from 1989 to 1994.

Carlo Rubbia's pioneering work on the proton–antiproton project led to the discovery of the [W](#) and [Z](#) bosons in 1983, earning CERN its first Nobel Prize in [1984](#). This discovery not only advanced understanding of the fundamental forces of nature through innovative detector and accelerator developments, but also established CERN as a global leader in particle physics, paving the way for the [Large Hadron Collider \(LHC\)](#). Now is your chance to hear first-hand about his remarkable contributions to physics, science and society. On Friday 18 October, from 2 p.m. in the Main Auditorium, distinguished scientists will present and discuss his major achievements, including his role in the development of the Standard Model, breakthroughs in detector and accelerator technology, his impact on CERN's history, including his time as Director-General, his proposals for innovative sustainable energy technologies and his influential role in fostering international cooperation. There will then be the chance to raise a glass at a drinks reception.

[Register now](#) to attend. The event is open to all and registration is required for organisational purposes and to issue CERN access cards for non-CERN attendees. Text or video messages can be sent to audiovisual productions.service@cern.ch, who will make them available for the event.

Detector now collecting data in Japan after upgrade at CERN

The ND280 detector, an essential part of the [T2K \(Tokai to Kamioka\)](#) neutrino experiment in Japan, has undergone a significant upgrade to enhance its ability to precisely measure neutrino interactions. T2K aims to study neutrinos to improve our understanding of [matter-antimatter asymmetry](#) in the universe.

At the start of its second data collection phase in June 2024, the upgraded near detector ND280, which contains [multiple subdetectors](#) assembled and tested at CERN, was used for the first time. This marks the end of an extraordinary journey

that started in 2017 as an R&D project between T2K and CERN's Neutrino Platform.

After all the parts arrived in Japan, the detector was tested on the surface before being lowered it into its final position about 50 metres below ground in late 2023. "ND280 passed its first crucial test with flying colours in December 2023," says Stefano Levorato, researcher at INFN. "It was so incredible to see the first event displays with the neutrino tracks."

Read more on the [BE-EA website](#) and in the [CERN Courier](#).
Kristiane Bernhard-Novotny

Computer Security: AI attacks

What a cool, new, artificially intelligent world! There's [ChatGPT](#) to write your love letter, CV or paper. [Loudly](#) or [Udio](#) to compose your own music. [DALL-E](#) to produce "exceptionally accurate images" of your own creation. And eventually, in the future, much-improved AI might allow you to rewatch your favourite movie with you starring in the main role. Stefan becomes Ethan, James or Obi-Wan! Copyright issues aside, the future has plenty of ideas in store. And there are also AI and LLMs (large language models) "[to remarkably increase the efficiency, sensitivity and modelling of CERN experiments](#)". Or to help with the [stock market](#) and portfolio management. But with every good idea comes the dark side. AI used for [military applications](#)*. Or for [cyberattacks](#). So, while AI – [coming with its own computer security problems](#), too – might make daily life easier, it will also open up a new series of attack vectors.

The obvious first vector is that AI has started to be misused to create "better" phishing emails that are even harder to spot by our automatisms and by you because their content makes more sense, their language is more accurate and the number of mistakes and typos is minimal. The cat-and-mouse game between (now AI-driven) attacks and our (surely AI-driven) spam and anti-malware filtering is entering a new [boss level](#). Good AI vs bad AI. Secondly, researchers and security companies have already started to use AI and LLMs to automatically identify vulnerabilities in software and online systems. Helping both friendly hackers

and malicious attackers, AI can be used to probe deep into software applications, identify a flaw and either notify everyone of the need to fix it or exploit that zero-day vulnerability and start compromising any system using that software. With well-trained AI, we can all improve our code-base (by the way, do you use our [GitLab security scanners](#)?) and avoid the daily blunder of bugs and typing mistakes leading to misconfigurations, buffer overflows or remotely exploitable vulnerabilities. We can [scan CERN's internet presence](#) for existing vulnerabilities. But so can the attackers. And they do, more and more efficiently scanning CERN's web servers and web services. AI attacks, the next level. And this is only the beginning. As in the past with "worms" – malware that automatically spreads – malicious AI can start to evolve while scouting for new systems. It learns (the intention of AI) and strikes harder on the second or third pass. Unfortunately, attacking AI has, like any other attacker, the advantage of time and the element of surprise. (AI) defence is no mean feat. The spiral of offence versus defence just took another turn. All we can do is to remain vigilant and careful and to [enforce our defence-in-depth strategy](#).

*Remember [Skynet](#), anyone?

Computer Security team

Official News

Members of the personnel shall be deemed to have taken note of official news published for the CERN community. Reproduction of all or part of this information by persons or institutions external to the Organization requires the prior approval of the CERN Management. The full list of official news is available here: <https://cern.ch/official-news>.

Authorisation to operate CERN lifting equipment

Modification and automation of the EDH request procedure

From 1 October 2024, the operation of lifting equipment (overhead travelling cranes, forklift trucks and elevated work platforms) as well as electrical tractors (RTL-PEFRA) will be subject to a new authorisation procedure in the EDH system. This new procedure, which has been brought into line with the rules in force ([SSI-M-1-2](#), [SSI-M-1-6](#), [SSI-M-1-7](#)), has been created in consultation with the managers of the fleet of lifting equipment and RTL-PEFRA. Henceforth, a request to a CERN manager of lifting equipment or RTL-PEFRA will be subject to prior approval by the hierarchy and proof of the user's physical fitness to operate the equipment.

Anyone needing to use lifting equipment or RTL-PEFRA as part of their work at CERN must submit a [Request to use lifting equipment/ RTL-PEFRA](#) in EDH (form available as of 1 October), attaching to it all the requested supporting documents. These supporting documents can vary according to the status of the requestor (employed or associated

member of the personnel or contractor) and the type of equipment concerned. These supporting documents will be checked during the authorisation process. The duration of validity of the authorisation will be aligned de facto with the duration of validity of the medical certificate, where applicable, and will under no circumstances exceed two years for lifting equipment and five years for RTL-PEFRA. The new procedure also gives each EDH signatory the right to revoke all or part of the issued authorisation if the authorisation conditions are no longer fulfilled. In such cases, the requestor will be automatically informed of the revocation. The detailed procedures applicable to employed members of the personnel (MPE) on the one hand, and to the associated members of the personnel (MPA), Users and contractors on the other hand, are available at [this link](#).

HSE unit

Voluntary insurance available for staff members, fellows and graduates

Two types of optional insurance are available, on a voluntary basis, to CERN staff members, fellows and graduates.

1. Loss of earnings insurance:

In accordance with Article R II 4.13 of the Staff Regulations, when an employed member of the personnel (MPE) accumulates more than 12 months of sick leave due to an illness or accident of non-occupational origin within a 36-month period, the MPE's remuneration is reduced to two-thirds from the 13th to the 30th month of absence, and to zero from the 31st to the 36th month of absence.

To enable MPEs to protect themselves from the financial consequences of such a reduction in their remuneration, CERN has signed a specific collective loss of earnings insurance contract with

an insurance company (UNIQA) offering competitive rates. This insurance has the advantage that, once it has been taken out, it does not require the policyholder to complete any further administrative procedures, as the Salaries Office manages the continuation of the remuneration entirely. Full details about this insurance and the procedure for signing up for it are available in the [Admin E-Guide](#).

2. Life insurance:

CERN has also signed a group life insurance contract with the Swiss insurance company HELVETIA.

MPEs who take up this insurance, which is available for an attractive premium, can select the total lump sum payable in the event of death as

well as designating the beneficiaries who will receive the lump sum in the event of their death. As with the loss of earnings insurance, the administrative procedure is handled by the Salaries Office. Staff members will be able to keep this insurance cover, under certain conditions, when they become members of the CERN pension fund.

Full details about this insurance and the procedure for signing up for it are available in the [Admin e-guide](#).

Further information regarding these two insurances is available from the FAP department's Personnel Accounting section (service-personnel-accounting@cern.ch).

FAP department

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

24 Sep 11:00 | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN
[The Voice of Science – Meet Dr. Katie Mack \(AstroKatie\)](#)

24 Sep 18:15 | At CERN | Meyrin site | CERN Running Club | EN & FR
[41st CERN Road Race 2024](#)

27 Sep 9:00 | Knowledge Sharing | 160/1-009 | Diversity and Inclusion Office | EN
["Je Me Souviens" Celebrating Diversity & Inclusion at CERN@70](#)

1 Oct 15:00 | Knowledge Sharing | Online | CERN70 | EN
[CERN70 official ceremony: Inspiring the future](#) (more [CERN70](#) events) *We invite our CERN community to follow the event online. Attendance by invitation only.*

2 Oct 11:00 | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN
[Meet the author of "The Beauty of Falling – A Life in Pursuit of Gravity"](#)

4 Oct 18:00 | Knowledge Sharing | CERN Science Gateway | Public events | FR
[Partage ta science – 2024](#)

10 Oct 18:00 | Knowledge Sharing | Online | CERN Alumni | EN
[News from the Lab | CERN openlab](#)

11 Oct 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN
[L&D Micro-talk 12 – Demystifying creativity](#)

15 Oct 17:30 | Knowledge Sharing | Council Chamber | Staff Association | FR
[« Qui est légitime à parler du temps », par Etienne Klein](#)

17 Oct 13:30 | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN
[Meet the author of "Relic Gravitons"](#)

18 Oct 14:00 | Knowledge Sharing | Main Auditorium | Commemorations | EN
[Symposium to celebrate Carlo Rubbia's 90th birthday](#)

Registration now open

27 Nov–6 Dec | Experiments | CERN | DRD1 | EN
[DRD1 Gaseous Detectors School](#)

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events

Indico also shows ALL events happening [today, this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

Sign your child up for the *Futur en tous genres* programme and share your passion for your work!

On Thursday, 14 November 2024, join us for the CERN leg of Switzerland's *Futur en tous genres* programme, an initiative that aims to broaden the professional horizons of young people beyond gender stereotypes.

The event, organised in collaboration with the Swiss Department of Public Education, Training and Youth (DIP), will take place at CERN, in French, from 9 a.m. to 12.15 p.m. We will invite the children to explore various professional fields, encouraging them to reflect freely and without bias on their future careers.

Who can attend? The event is open to the children of CERN personnel who go to school in Switzerland,

in the cantons of Geneva (9e), Vaud (7H to 9H), et Valais (8H).

The number of places is limited. Please visit this link for further information and to sign your child up for the event: [Registration for Futur en tous genres 2024](#).

This year, we also want to invite members of our community to present their jobs. If you are passionate about your work and want to inspire the next generation, this is a golden opportunity to make a difference. The presentations will take place in the Globe from 11.30 a.m. to 12.15 p.m. If you wish to take part, please contact education.locale@cern.ch

Closure of the tunnel linking the CERN sites on 23 and 24 September

The **tunnel linking the CERN sites** will undergo renovation works from 20 September 2024 for two weeks. It **will be completely closed to vehicles on 23 and 24 September all day**. Note that the access to **Route Schrödinger** will also be **closed** for the Entrance E direction until 7 October inclusive. Traffic control signage and barriers will be installed on the roads giving access to the tunnel. Apart from this temporary closure, traffic during the renovation period is expected to be minimally impacted. The tunnel linking the CERN sites will

remain open from 7:00 a.m. to 6:00 p.m. under the supervision of a security guard. Transit through that gate (which is a border crossing) will still require specific transit authorisation ([ADaMS](#)). We also invite you to take note of the [traffic restrictions surrounding the 1 October event CERN70: Inspiring the future](#). Apologies for the inconvenience.

SCE department

The smart(phone) way to celebrate CERN70

As CERN's 70th birthday approaches, why not extend the celebrations to your phone?

You can now download exclusive CERN70 wallpapers for your phone screen from the CERN design guidelines website. Watch [the video](#) to find out more.

Why not extend the celebrations with social media? You can now use special filters to showcase your own CERN70 stories on Instagram. [Click here](#) to find out more.

And if you haven't done so already, [download](#) the CERN CampusApp for all the latest CERN70 news and events.